Research Methodology

Week 12 Assignment: Questions, Answers and Explanations

Week 12: Hypothesis Generation, Thinking Habits and Metacognition

NPTEL course noc26_ge79 (SWAYAM), 10 questions, 1 point each

Correct answers are the NPTEL portal's accepted answers. For every option the table gives a verdict and the reason, and each question ends with a short explanation of the correct answer. This week covers research versus course-based education, hypotheses and falsifiability, Bloom's taxonomy and creativity. You lost marks only on Question 9.

Answer key at a glance

Question	Type	Correct option(s)	Your portal score
1	Select all that apply	**A, C**	1 (full marks)
2	Select all that apply	**A, C, D**	1 (full marks)
3	Select all that apply	**A, B**	1 (full marks)
4	Select all that apply	**A, C, D**	1 (full marks)
5	Select all that apply	**A, C**	1 (full marks)
6	Select all that apply	**A, B, D**	1 (full marks)
7	Select all that apply	**A, B**	1 (full marks)
8	Select all that apply	**A, B, D**	1 (full marks)
9	Select all that apply	**A, B, D**	0.67 (partial: two of the three correct options chosen)
10	Select all that apply	**A, B**	1 (full marks)

Questions with explanations

Question 1 of 10

Which of the following statements correctly distinguish course-based education from research-based education as discussed in the lecture?

Select all that apply.

Option	Statement	Verdict	Reason
A	**Course-based education primarily deals with an existing body of knowledge, whereas PhD research is expected to create new knowledge.**	**Correct**	A course teaches what is known; a PhD must add something new.
B	A PhD programme follows a common prescribed syllabus and curriculum in the same way as a course-based degree.	**Wrong**	A PhD has no common prescribed syllabus.
C	**Course-based education generally has a prescribed syllabus and curriculum, whereas research-based education does not simply follow a fixed syllabus.**	**Correct**	Research follows the problem, not a fixed syllabus.
D	Knowledge acquired in course-based education and new knowledge generated through PhD research are evaluated in exactly the same way.	**Wrong**	Exams test known material; research is judged by its originality and quality.

Correct answer: A and C.

Explanation: A course teaches an existing body of knowledge under a syllabus, while PhD research must create new knowledge and does not follow a fixed syllabus. The two are not evaluated in the same way.

Question 2 of 10

A research scholar is beginning a PhD and expects the research guide to provide a complete step-by-step procedure for the entire research programme. Which responses are consistent with the lecture?

Select all that apply.

Option	Statement	Verdict	Reason
A	**The scholar should recognize that PhD research involves substantially self-directed work.**	**Correct**	Research is largely self-directed.
B	The absence of a fixed syllabus means that the scholar does not need guidance from the research guide.	**Wrong**	No syllabus does not mean no guidance.
C	**The research guide can provide guidance and support when ambiguity or difficulty arises.**	**Correct**	The guide helps when the path is unclear.
D	**The scholar is expected to develop the ability to decide how the research should progress.**	**Correct**	Deciding the next step is part of learning to do research.

Correct answer: A and C and D.

Explanation: PhD research is largely self-directed, and the scholar must learn to decide the next step, while the guide supports when things are unclear. A step-by-step procedure is not provided.

Question 3 of 10

A researcher proposes a hypothesis based on existing knowledge and experience. Which conditions, according to the lecture, are relevant to treating it as a scientific hypothesis?

Select all that apply.

Option	Statement	Verdict	Reason
A	**It should be an educated prediction rather than a statement already established as true.**	**Correct**	A hypothesis is an informed guess that is still to be tested.
B	**It should allow a possibility of being disproved.**	**Correct**	A scientific hypothesis must be falsifiable.
C	It should be protected from contradictory observations by adding explanations that cannot be tested.	**Wrong**	This makes the hypothesis unfalsifiable.
D	Its truth must be established before experiments or data collection begin.	**Wrong**	If it were already established there would be nothing to test.

Correct answer: A and B.

Explanation: A scientific hypothesis is an educated prediction that can be tested and could be shown false (falsifiability). Shielding it with untestable excuses, or requiring it to be proved beforehand, defeats the purpose.

Question 4 of 10

Consider the hypothesis: “All swans are white.” Based on the falsifiability example in the lecture, which statements are correct?

Select all that apply.

Option	Statement	Verdict	Reason
A	**Finding a black swan would provide an observation capable of disproving the hypothesis.**	**Correct**	A single black swan contradicts "all swans are white".
B	It is necessary to observe every swan in the world before the hypothesis can be scientifically examined.	**Wrong**	It can be examined and tested without checking every swan.
C	**The possibility of disproving the hypothesis is central to the criterion discussed in the lecture.**	**Correct**	Falsifiability is the criterion.
D	**The criterion discussed emphasizes the possibility of disproof rather than establishing the hypothesis as permanently true.**	**Correct**	No number of white swans proves it, but one black swan disproves it.

Correct answer: A and C and D.

Explanation: Any number of white swans cannot prove that all swans are white, but one black swan disproves it. A scientific hypothesis is one that could be shown false, not one that is proved permanently true.

Question 5 of 10

A student starts a PhD after being given a broad research topic. According to the lecture, which actions are appropriate before arriving at a testable hypothesis?

Select all that apply.

Option	Statement	Verdict	Reason
A	**Begin by formulating relevant questions about the topic and considering why the topic is important.**	**Correct**	Questions give direction to the work.
B	Start an unrestricted literature survey immediately and use it as the only route for identifying the research problem.	**Wrong**	Unfocused reading as the only route is not appropriate.
C	**Use literature exploration and observations to refine the initial questions through repeated cycles.**	**Correct**	Questions are refined in rounds of reading and observing.
D	Formulate the final testable hypothesis immediately from the broad topic without further exploration.	**Wrong**	It is premature.

Correct answer: A and C.

Explanation: Start with questions about the topic and its importance, then refine them through repeated rounds of reading and observation. A final hypothesis straight from a broad topic is premature.

Question 6 of 10

In the obesity and eating-habit example, observations from the literature and study were used to identify an anomaly and develop a hypothesis. Which statements are supported by the lecture?

Select all that apply.

Option	Statement	Verdict	Reason
A	**An anomaly can arise when observations do not match a commonly assumed explanation.**	**Correct**	That mismatch is what an anomaly is.
B	**The observed influence of factors such as the amount of food offered can motivate further investigation.**	**Correct**	Such an unexpected effect is a lead worth investigating.
C	The hypothesis was formulated directly from the broad topic without literature survey or observations.	**Wrong**	It came from literature and observations.
D	**Identifying a missing link or contradiction can provide scope for further research and help generate a hypothesis.**	**Correct**	Gaps and contradictions suggest hypotheses.

Correct answer: A and B and D.

Explanation: The example shows a hypothesis coming from the literature and observations: something did not match the usual explanation, such as the effect of the amount of food offered, and that gap led to a hypothesis.

Question 7 of 10

According to the Revised Bloom's Taxonomy discussed in the lecture, which statements are correct?

Select all that apply.

Option	Statement	Verdict	Reason
A	**Remember, Understand and Apply are described as lower-order thinking levels.**	**Correct**	These are the three lower levels.
B	**Analyze, Evaluate and Create are described as higher-order thinking levels.**	**Correct**	These are the three higher levels.
C	Create is placed below Remember because generating a new idea generally requires less cognitive effort.	**Wrong**	Create is at the top because it needs the most thinking.
D	The taxonomy states that every individual experiences exactly the same difficulty across all six levels.	**Wrong**	The taxonomy says nothing like this.

Correct answer: A and B.

Explanation: The six levels run from Remember, Understand and Apply (lower order) to Analyze, Evaluate and Create (higher order). Create is at the top, and the taxonomy makes no claim that everyone finds all levels equally hard.

Question 8 of 10

A PhD scholar collects experimental data, examines unfamiliar patterns, considers alternative hypotheses, and develops a new research idea. Which Bloom's levels are most directly represented by these activities?

Select all that apply.

Option	Statement	Verdict	Reason
A	**Analyze**	**Correct**	Examining data and patterns is analysing.
B	**Evaluate**	**Correct**	Weighing alternative hypotheses is evaluating.
C	Remember	**Wrong**	Recalling facts is not the activity described.
D	**Create**	**Correct**	Developing a new research idea is creating.

Correct answer: A and B and D.

Explanation: Examining data and patterns is analysing, considering alternative hypotheses is evaluating and developing a new idea is creating. Remembering is not the activity described.

Question 9 of 10

Which of the following statements correctly reflect the lecture's discussion of the role of higher-order thinking in PhD research?

Select all that apply. *Your portal score: 0.67 (partial: two of the three correct options chosen).*

Option	Statement	Verdict	Reason
A	**All six levels of thinking may be used during research problem solving.**	**Correct**	Research uses every level, from recalling known results to creating new ones.
B	**The lecture identifies analysis as an important starting level in PhD research because researchers collect and analyze data.**	**Correct**	Research starts by collecting and analysing data.
C	The main intellectual focus of a PhD is memorizing information from the existing body of knowledge.	**Wrong**	Memorising is a lower-order activity and not the focus of a PhD.
D	**Evaluation can involve considering alternate hypotheses and attempting to validate them.**	**Correct**	Weighing and testing alternatives is evaluation.

Correct answer: A and B and D.

Explanation: Research draws on all six levels and starts with analysing data, and evaluation covers weighing and testing alternative hypotheses. The main focus of a PhD is higher-order thinking, not memorising.

Question 10 of 10

Which of the following statements are consistent with the lecture's discussion of creativity and functional fixedness?

Select all that apply.

Option	Statement	Verdict	Reason
A	**Creative thinking can be developed or improved through practice rather than being limited to an ability fixed at birth.**	**Correct**	Creativity is a skill that practice improves.
B	**Functional fixedness can cause a person to perceive an object only through its conventional purpose and can hinder lateral thinking.**	**Correct**	Seeing an object only for its usual use blocks new uses.
C	Complete knowledge of the existing body of knowledge guarantees the creation of something new.	**Wrong**	Knowing everything known does not guarantee anything new.
D	Functional fixedness helps researchers generate new approaches when facing unexpected or novel challenges.	**Wrong**	Functional fixedness is an obstacle to new approaches.

Correct answer: A and B.

Explanation: Creativity can be improved by practice. Functional fixedness is seeing an object only for its usual purpose (a brick only as a building block), which blocks lateral thinking, so it hinders new approaches.